

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283659

Kind Code

A1

Publication Date

September 11, 2025

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PORTABLE COOLER WITH ACTIVE TEMPERATURE CONTROL

Abstract

A portable cooler container with active temperature control system is provided. The active temperature control system is operated to heat or cool a chamber of a vessel to approach a temperature set point suitable for a medication stored in the cooler container.

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Family ID: 1000008619619

Appl. No.: 19/215188

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 17301932 20210419 PENDING child US 19215188

parent US continuation 16736558 20200107 parent-grant-document US 10989466 child US 17301932

us-provisional-application US 62791225 20190111

us-provisional-application US 62827636 20190401

Publication Classification

Int. Cl.: F25D31/00 (20060101); F25B21/02 (20060101); F25B21/04 (20060101)

U.S. Cl.:

Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57 and should be considered a part of this specification.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention is directed to a portable cooler (e.g., for medicine such as insulin, vaccines, epinephrine, etc.), and more particularly to a portable cooler with active temperature control.

Description of the Related Art

[0003] Certain medicine needs to be maintained at a certain temperature or temperature range to be effective (e.g., to maintain potency). Once potency of medicine (e.g., a vaccine, insulin, epinephrine) is lost, it cannot be restored, rendering the medicine ineffective and/or unusable. For example, injector pens are commonly used to deliver medication, such as epinephrine to counteract the effects of an allergic reaction (e.g., due to a peanut allergy, insect stings/bites, etc.). Users sometimes carry such medicine (e.g., medicine injector pens, cartridges for injector pens) with them (e.g., in a bag, purse, pocket, etc.) in the event they suffer an allergic reaction during the day. However, such medicine may be exposed to varying temperatures during the day (e.g., due to ambient temperature conditions, temperature conditions in the car, workplace, school, etc.), which can be outside the preferred temperature or temperature range for the medicine to be effective.

SUMMARY

[0004] Accordingly, there is a need for improved portable cooler designs (e.g., for storing and/or transporting medicine, such as epinephrine, vaccines, insulin, etc.) that can maintain the contents of the cooler at a desired temperature or temperature range. Additionally, there is a need for an improved portable cooler design with improved cold chain control and record keeping of the temperature history of the contents (e.g., medicine, such as epinephrine, vaccines, insulin, etc.) of the cooler (e.g., during storage and/or transport of the medicine, such as during a commute to work or school).

[0005] In accordance with one aspect, a portable cooler container (e.g., capsule) with active temperature control system is provided. The active temperature control system is operated to heat or cool a chamber of a vessel to approach a temperature set point suitable for a medication (e.g., epinephrine, insulin, vaccines, etc.) stored in the cooler container.

[0006] In accordance with another aspect, a portable cooler (or capsule) is provided that includes a temperature control system operable (e.g., automatically operable) to maintain the chamber of the cooler at a desired temperature or temperature range for a prolonged period of time. Optionally, the portable cooler is sized to house one or more containers (e.g., injector pens and/or cartridges for injector pens, vials, etc.). Optionally, the portable cooler automatically logs (e.g., stores on a memory of the cooler) and/or communicates data on one or more sensed parameters (e.g., of the temperature of the chamber, battery charge level, etc.) to a remote electronic device (e.g., remote computer, mobile electronic device such as a smartphone or tablet computer). Optionally, the portable cooler can automatically log and/or transmit the data to the remote electronic device (e.g., automatically in real time, periodically at set intervals, etc.).

[0007] In accordance with another aspect, a portable cooler container (e.g., capsule) with active temperature control is provided. The container comprises a container body having a chamber configured to receive and hold one or more containers (e.g., injector pens, cartridges for injector pens, vials, etc.), the chamber defined by a base and an inner peripheral wall of the container body. The container also comprises a temperature control system comprising one or more thermoelectric elements (e.g., Peltier elements) configured to actively heat or cool a heat sink component in thermal communication (e.g., in contact with) the one or more containers (e.g., medicine containers) in the chamber, and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the heat sink component and/or chamber to a predetermined temperature or temperature range.

[0008] Optionally, the container can include one or more batteries configured to provide power to one or both of the circuitry and the one or more thermoelectric elements.

[0009] Optionally, the circuitry is further configured to wirelessly communicate with a cloud-based data storage system (e.g., remote server) or a remote electronic device (e.g., smartphone, tablet computer, laptop computer, desktop computer).

[0010] Optionally, the container includes a first heat sink in thermal communication with the chamber, the first sink being selectively thermally coupled to the one or more thermoelectric elements. Optionally, the first heat sink can removably extend into the chamber of the container and one or more containers (e.g., medicine containers, such as injector pens, cartridges for injector pens, vials, etc.) can releasably couple to the first heat sink (e.g., to one or more clip portions or slots of the first heat sink) so that the one or more containers are disposed in the chamber.

[0011] Optionally, the container includes a second heat sink in communication with the one or more thermoelectric elements (TECs), such that the one or more TECs are disposed between the first heat sink and the second heat sink.

[0012] Optionally, the second heat sink is in thermal communication with a fan operable to draw heat from the second heat sink.

[0013] In one implementation, such as where the ambient temperature is above the predetermined temperature or temperature range, the temperature control system is operable to draw heat from the first heat sink (and draw heat from the chamber), which transfers said heat to the one or more TECs, which transfer said heat to the second heat sink, where the optional fan dissipates heat from the second heat sink. The temperature control system can in this manner cool the first heat sink (and the chamber), thereby cooling the containers (e.g., medicine containers) in the chamber toward the predetermined temperature or temperature range.

[0014] In another implementation, such as where the ambient temperature is below the predetermined temperature or temperature range, the temperature control system is operable to add heat to the first heat sink (and add heat to the chamber), which transfers said heat from the one or more TECs. The temperature control system can in this matter heat the first heat sink (and the chamber), thereby heating the containers (e.g., medicine containers) in the chamber toward the predetermined temperature or temperature range.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 is a schematic view of one embodiment of a cooler container.

[0016] FIG. 2 is a schematic view of the cooler container of FIG. 1 on one embodiment of a charging base.

[0017] FIG. 3 is a partial view of the cooler container of FIG. 1, with a lid detached from the vessel of the cooler container, with three injector pens and/or cartridges coupled to the heat sink attached to the lid.

[0018] FIG. 4 is a schematic cross-sectional view of the cooler container of FIG. 1.

[0019] FIG. 5 is a schematic view of the cooler container of FIG. 1 in communication with a remote electronic device.

[0020] FIG. 6 is a schematic view of the cooler container of FIG. 1 and another embodiment of a charging base.

[0021] FIG. 7 is a schematic cross-sectional view of another embodiment of a cooler container.

[0022] FIG. 8 is a schematic cross-sectional view of a vessel of the cooler container of FIG. 7 without the lid.

[0023] FIG. 9 is a schematic block diagram showing communication between the cooler container and a remote electronic device.

[0024] FIG. 10A is a schematic partial perspective view of another cooler container.

[0025] FIG. 10B is a schematic cross-sectional view of the cooler container of FIG. 10A.

[0026] FIG. 11A is a schematic partial perspective view of another cooler container.

[0027] FIG. 11B is a schematic cross-sectional view of the cooler container of FIG. 11A.

[0028] FIG. 11C is a schematic cross-sectional view of the cooler container in FIG. 11A.

[0029] FIGS. 12A-12C is a schematic cross-sectional view of another cooler container.

[0030] FIG. 13 is a schematic partial cross-sectional view of a portion of another cooler container.

[0031] FIGS. 14A-14B are a schematic partial cross-sectional view of another cooler container.

[0032] FIG. 15 is a schematic partial cross-sectional view of another cooler container.

[0033] FIG. 16 shows a schematic perspective view of another cooler container and an exploded view of a capsule for use with the container.

[0034] FIG. 16A shows a schematic cross-sectional view of a capsule for use with the cooler container of FIG. 16.

[0035] FIG. 16B shows a schematic cross-sectional view of another capsule for use with cooler container of FIG. 16.

[0036] FIG. 16C shows an enlarged cross-sectional view of a portion of the capsule in FIG. 16B.

[0037] FIG. 17 shows a schematic perspective view of another cooler container.

[0038] FIG. 17A shows a schematic perspective view of a capsule for use with the cooler container of FIG. 17.

[0039] FIG. 17B shows a schematic cross-sectional view of the capsule in FIG. 17A for use with the cooler container of FIG. 17.

[0040] FIG. 18 shows a schematic perspective view of another cooler container.

[0041] FIG. 18A shows a schematic view of an injector pen for use with cartridges taken from the cooler container of FIG. 18.

[0042] FIG. 18B shows a schematic partial view of a cartridge from the cooler container of FIG. 18 loaded into an injector pen.

[0043] FIG. 19A shows a schematic perspective view of a cooler container.

[0044] FIG. 19B is a is a schematic block diagram showing electronics in the cooler container associated with operation of the display screen of the cooler container.

[0045] FIGS. 20A-20B show block diagrams of a method for operating the cooler container of FIG. 19A.

[0046] FIGS. 21A-21D are schematic user interfaces for an electronic device for use with a cooler container.

[0047] FIG. 22A is a schematic longitudinal cross-sectional view of a cooler container.

[0048] FIG. 22B is a schematic transverse cross-sectional view of the cooler container in FIG. 22A.

DETAILED DESCRIPTION

[0049] FIGS. 1-8 show a container system 100 (e.g., capsule container) that includes a cooling system 200. Optionally, the container system 100 has a container vessel 120 that is optionally cylindrical and symmetrical about a longitudinal axis Z, and one of ordinary skill in the art will recognize that the features shown in cross-section in FIGS. 4, 7 and 8 defined by rotating them

about the axis Z to define the features of the container **100** and cooling system **200**.

[0050] The container vessel **120** is optionally a cooler with active temperature control provided by the cooling system **200** to cool the contents of the container vessel **120** and/or maintain the contents of the vessel **120** in a cooled or chilled state. Optionally, the vessel **120** can hold therein one or more (e.g., a plurality of) separate containers **150** (e.g., medicine containers, such as injector pens, vials, cartridges (such as for injector pens), etc.). Optionally, the one or more (e.g., plurality of) separate containers **150** that can be inserted into the container vessel **120** can contain a medication or medicine (e.g., epinephrine, insulin, vaccines, etc.).

[0051] The container vessel **120** has an outer wall **121** that extends between a proximal end **122** that has an opening **123** and a distal end **124** having a base **125**. The opening **123** is selectively closed by a lid **L** removably attached to the proximal end **122**. As shown in FIG. 4, the vessel **120** has an inner wall **126A** and a base wall **126B** that together define an open chamber **126** that can receive and hold contents to be cooled therein (e.g., medicine containers, such as one or more vials, cartridges, injector pens, etc.). The vessel **120** can optionally have an intermediate wall **126C** spaced about the inner wall **126A** and base wall **126B**, such that the intermediate wall **126C** is at least partially disposed between the outer wall **121** and the inner wall **126A**. The intermediate wall **126C** is spaced apart from the inner wall **126A** and base wall **126B** so as to define a gap **G** between the intermediate wall **126C** and the inner wall **126A** and base wall **126B**. The gap **G** can optionally be under vacuum so that the inner wall **126A** and base wall **126B** are vacuum insulated relative to the intermediate wall **126C** and the outer wall **121** of the vessel **120**.

[0052] Optionally, one or more of the inner wall **126A**, intermediate wall **126B** and outer wall **121** can be made of metal (e.g., stainless steel). In one implementation, the inner wall **126A**, base wall **126B** and intermediate wall **126C** are made of metal (e.g., stainless steel). In another implementation, one or more portions (e.g., outer wall **121**, intermediate wall **126C** and/or inner wall **126A**) of the vessel **120** can be made of plastic.

[0053] The vessel **120** has a cavity **127** between the base wall **126B** and a bottom **275** of the vessel **120**. The cavity **127** can optionally house one or more batteries **277**, and one or more printed circuit boards (PCBA) **278** with circuitry that controls the cooling system **200**. In one implementation, the cavity **127** can optionally house a power button or switch actuatable by a user through the bottom of the vessel **275**, as further described below. Optionally, the bottom **275** defines at least a portion of an end cap **279** attached to the outer wall **121**. Optionally, the end cap **279** is removable to access the electronics in the cavity **127** (e.g., to replace the one or more batteries **277**, perform maintenance on the electronics, such as the PCBA **278**, etc.). The power button or switch is accessible by a user (e.g., can be pressed to turn on the cooling system **200**, pressed to turn off the cooling system **200**, pressed to pair the cooling system **200** with a mobile electronic device, etc.). Optionally, the power switch can be located generally at the center of the end cap **279** (e.g., so that it aligns/extends along the longitudinal axis Z of the vessel **120**).

[0054] With continued reference to FIGS. 1-8, the cooling system **200** is optionally at least partially housed in the lid **L** that releasably closes the opening **123** of the vessel **120**. In one implementation, the lid **L** can releasably couple to the vessel **120** via one or more magnets in the lid **L** and/or in the vessel **120**. In other implementations, the lid **L** can releasably couple to the vessel **120** via other suitable mechanisms (e.g., threaded connection, key-slot connection, press-fit connection, etc.)

[0055] In one implementation, the cooling system **200** can include a first heat sink (cold side heat sink) **210** in thermal communication with one or more thermoelectric elements (TECs) **220**, such as Peltier element(s), and can be in thermal communication with the chamber **126** of the vessel **120** (e.g., via contact with the inner wall **126A**, via conduction with air in the chamber **126**, etc.). Optionally, cooling system **200** can include an insulator member (e.g., insulation material) disposed between the first heat sink **210** and a second heat sink **230**.

[0056] With continued reference to FIGS. 1-8, the TEC **220** is selectively operated (e.g., by the circuitry **278**) to draw heat from the first heat sink (e.g., cold-side heat sink) **210** and transfer it to

the second heat sink (hot-side heat sink) **230**. A fan **280** is selectively operable to draw air into the lid **L** to dissipate heat from the second heat sink **230**, thereby allowing the TEC **220** to draw further heat from the first heat sink **210**, and thereby draw heat from the chamber **126**. During operation of the fan **280**, intake air flow F_i is drawn through one or more intake vents **203** (having one or more openings **203A**) in the lid **L** and over the second heat sink **230** (where the air flow removes heat from the second heat sink **230**), after which the exhaust air flow F_o flows out of one or more exhaust vents **205** (having one or more openings **205A**) in the lid **L**.

[0057] As shown in FIG. 4, the chamber **126** optionally receives and holds one or more (e.g., a plurality of) containers **150** (e.g., medicine containers, such as injector pens or cartridges for injector pens, vials, etc.). The first heat sink **210** can define one or more slots **211** that can receive and hold (e.g., resiliently receive and hold) one or more of the containers **150**. Therefore, during operation of the cooling system **200**, the first heat sink **210** is cooled, which thereby cools the one or more containers **150** coupled to the heat sink **210**. In one implementation, the first heat sink **210** can be made of aluminum. However, the first heat sink **210** can be made of other suitable materials (e.g., metals with high thermal conductivity).

[0058] The electronics (e.g., PCBA **278**, batteries **277**) can electrically communicate with the fan **280** and TEC **220** in the lid **L** via one or more electrical contacts (e.g., electrical contact pads, Pogo pins) **281** in the lid **L** (e.g., downward facing electrical contacts, contact pads or Pogo pins) that contact one or more electrical contacts (e.g., Pogo pins, electrical contact pads) **282** in the portion of the vessel **120** (e.g., upward facing electrical contacts, contact pads or Pogo pins) that engages the lid **L**. Advantageously, the electrical contacts **281**, **282** facilitate the coupling of the lid **L** to the vessel **120**, **120'** in the correct orientation (alignment) to allow the contact between the electrical contacts **282**, **281** (e.g., provide a clocking feature). As shown in FIG. 3, the one or more electrical contacts **282** can be a set of eight contacts **282** that interface with an equal number of electrical contacts **281** in the lid **L**. However, different number of electrical contacts **282**, **281** are possible. Electrical leads can extend from the PCBA **278** along the side of the vessel **120** (e.g., between the outer wall **121** and the intermediate wall **126C**) to the electrical contacts **282**. Accordingly, power can be provided from the batteries **277** to the TEC **220** and/or fan **280**, and the circuitry (e.g., in or on the PCBA **278**) can control the operation of the TEC **220** and/or fan **280**, via one or more of the electrical contacts **281**, **282** when the lid **L** is coupled to the vessel **120**. As further discussed below, the lid **L** can have one or more sensors, and such sensors can communicate with the circuitry (e.g., in or on the PCBA **278**) via one or more of the electrical contacts **281**, **282**.

[0059] FIGS. 7-8 schematically illustrate the container system **100** with the cooling system **200** and a vessel **120'**. The cooling system **200** is similar to the cooling system **200** in the container **100** of FIGS. 1-7. Some of the features of the vessel **120'** are similar to features in the vessel **120** in FIGS. 1-7. Thus, references numerals used to designate the various components of the vessel **120'** are identical to those used for identifying the corresponding components of the vessel **120** in FIGS. 1-7, except that a "'" is added to the numerical identifier. Therefore, the structure and description for the various components of the cooling system **200** and vessel **120** in FIGS. 1-7 are understood to also apply to the corresponding components of the cooling system **200** and vessel **120'** in FIGS. 7-8, except as described below.

[0060] As shown in FIGS. 7-8, the vessel **120'** includes a cylindrical chamber wall **126D'** that defines the chamber **126'** and is spaced inward (e.g., toward the center of the chamber **126**) of the inner wall **126A'** and the base wall **126B'** so as to define a gap G_2' between the chamber wall **126D'** and the inner wall **126A'** and base wall **126B'**. optionally, the gap G_2' is filled with a phase change material (PCM) **130'**. In one implementation, the phase change material **130'** can be a solid-fluid PCM. In another implementation, the phase change material **130'** can be a solid-solid PCM. The PCM **130'** advantageously can passively absorb and release energy. Examples of possible PCM materials are water (which can transition to ice when cooled below the freezing temperature), organic PCMs (e.g., bio based or Paraffin, or carbohydrate and lipid derived), inorganic PCMs

(e.g., salt hydrates), and inorganic eutectics materials. However, the PCM **130'** can be any thermal mass that can store and release energy.

[0061] In operation, the cooling system **200** can be operated to cool the heat sink **210** to cool the one or more containers **150** that are coupled to the heat sink **210**, and to also cool the chamber **126'**. The cooling system **200** can optionally also cool the PCM **130'** (e.g., via the chamber wall **126D'**). In one implementation, the cooling system **200** optionally cools the PCM **130'** via conduction (e.g., contact) between at least a portion of the heat sink **210** and at least a portion of the chamber wall **126D'** (e.g., near the opening **123'** of the vessel **120'**). In another implementation, the cooling system **200** optionally cools the PCM **130'** via conduction through the air in the chamber **126'** between the heat sink **210** and the chamber wall **126D'**.

[0062] Advantageously, the PCM **130'** operates as a secondary (e.g., backup) cooling source for the chamber **126'** and/or the containers **150'** (e.g., medicine containers, such as injector pens, cartridges for injector pens, vials, etc.) disposed in the chamber **126'**. For example, if the one or more intake vents **203** are partially (or fully) blocked (e.g., because they are up against a surface of a handbag, backpack, suitcase, during travel; due to dust accumulation in the vent openings **203A**) or if the cooling system **200** is not operating effectively due to low charge in the one or more batteries **277**, the PCM **130'** can maintain the one or more containers **150** (e.g., injector pens, cartridges for injector pens, vials, etc.) in a cooled state until the vents **203** are unblocked/unclogged, one or more batteries **277** are charged, etc. Though the phase change material **130'** is described in connection with the chamber **126'** and container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** one of skill in the art will recognize that it can also be applied to all the other implementations discussed herein for the chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126H**, **126I**, **126J**, **126K** and container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**.

[0063] The container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** disclosed herein can optionally communicate (e.g., one-way communication, two-way communication) with one or more remote electronic devices (e.g., mobile phone, tablet computer, desktop computer, remote server) **600**, via one or both of a wired or wireless connection (e.g., 802.11b, 802.11a, 802.11g, 802.11n standards, etc.). Optionally, the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** can communicate with the remote electronic device **600** via an app (mobile application software) that is optionally downloaded (e.g., from the cloud) onto the remote electronic device **600**. The app can provide one or more graphical user interface screens **610** via which the remote electronic device **600** can display one or more data received from the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** and/or information transmitted from the remote electronic device **600** to the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**. Optionally, a user can provide instructions to the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** via the one or more of the graphical user interface screens **610** on the remote electronic device **600**.

[0064] In one variation, the graphical user interface (GUI) screen **610** can provide one or more temperature presets corresponding to one or more particular medications (e.g., epinephrine/adrenaline for allergic reactions, insulin, vaccines, etc.). The GUI screen **610** can optionally allow the turning on and off of the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L**. The GUI screen **610** can optionally allow the setting of the control temperature to which one or both of the first heat sink **210** and the chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126H**, **126I**, **126J**, **126K**, **126L** in the container **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** is cooled by the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L**.

[0065] In another variation, the graphical user interface (GUI) screen **610** can provide a dashboard display of one or more parameters of the container **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** (e.g., ambient temperature, internal temperature in the chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126H**, **126I**, **126J**, **126K**, **126L** temperature of the first heat sink **210**,

temperature of the one or more batteries **277**, etc.). The GUI screen **610** can optionally provide an indication (e.g., display) of power supply left in the one or more batteries **277** (e.g., % of life left, time remaining before battery power drains completely). Optionally, the GUI screen **610** can also include information (e.g., a display) of how many of the slots or receptacles **211** in the first heat sink **210** are occupied (e.g., by containers **150**, **150J**). Optionally, the GUI screen **610** can also include information on the contents of the container **100** (e.g., medication type, such as insulin, or disease medication is meant to treat, such as Hepatitis, etc.) and/or information (e.g., name, identification no., contact info) for the individual to whom the container **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** belongs.

[0066] In another variation, the GUI screen **610** can include one or more notifications provided to the user of the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** disclosed herein, including alerts on battery power available, alerts on ambient temperature effect on operation of container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** alert on temperature of the first heat sink **210**, alert on temperature of the chamber **126**, **126'**, **126E**, **126F**, **126G**, **126H**, **126I**, **126J**, **126K**, **126L** alert on low air flow through the intake vent **203** and/or exhaust vent **205** indicating they may be blocked/clogged, etc. One of skill in the art will recognize that the app can provide the plurality of GUI screens **610** to the user, allowing the user to swipe between the different screens. Optionally, as discussed further below, the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K** can communicate information, such as temperature history of the chamber **126**, **126'**, **126E**, **126F**, **126G**, **126H**, **126I**, **126J**, **126K**, **126L** temperature history of the first heat sink **210** and/or chamber **126**, **126'**, **126E**, **126F**, **126G**, **126H**, **126I**, **126J**, **126K**, **126L** that generally corresponds to the temperature of the containers **150**, **150J**, temperature of the container **150**, **150J** from a temperature sensor on the container **150**, **150J**, power level history of the batteries **277**, ambient temperature history, etc. to one or more of a) an RFID tag on the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** that can later be read (e.g., at the delivery location), b) to a remote electronic device (e.g., a mobile electronic device such as a smartphone or tablet computer or laptop computer or desktop computer), including wirelessly (e.g., via WiFi 802.11, BLUETOOTH®, or other RF communication), and c) to the cloud (e.g., to a cloud-based data storage system or server) including wirelessly (e.g., via WiFi 802.11, BLUETOOTH®, or other RF communication). Such communication can occur on a periodic basis (e.g., every hour; on a continuous basis in real time, etc.). Once stored on the RFID tag or remote electronic device or cloud, such information can be accessed via one or more remote electronic devices (e.g., via a dashboard on a smart phone, tablet computer, laptop computer, desktop computer, etc.). Additionally, or alternatively, the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** can store in a memory (e.g., part of the electronics in the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**) information, such as temperature history of the chamber **126**, **126'**, **126E**, **126F**, **126G**, **126H**, **126I**, **126J**, **126K**, **126L** temperature history of the first heat sink **210**, power level history of the batteries **277**, ambient temperature history, etc., which can be accessed from the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** by the user via a wired or wireless connection (e.g., via the remote electronic device **600**).

[0067] With reference to FIGS. 1-9, the body **120** of the container **100** can optionally have a visual display on the outer wall **121** of the body **120**. The visual display can optionally display one or more of the temperature in the chamber **126**, **126'**, the temperature of the first heat sink **210**, the ambient temperature, a charge level or percentage for the one or more batteries **277**, and amount of time left before recharging of the batteries **277** is needed, etc. The visual display can optionally include a user interface (e.g., pressure sensitive buttons, capacitance touch buttons, etc.) to adjust (up or down) the temperature preset at which the cooling system **200** is to cool the chamber **126**, **126'**. Accordingly, the operation of the container **100** (e.g., of the cooling system **200**) can be selected via the visual display and user interface on a surface of the container **100**. Optionally, the

visual display can include one or more hidden-til-lit LEDs. Optionally, the visual display can include an electronic ink (e-ink) display. In one variation, the container **100** can optionally include a hidden-til-lit LED **140** that can selectively illuminate (e.g., to indicate one or more operating functions of the container **100**, such as to indicate that the cooling system **200** is in operation). The LED **140** can optionally be a multi-color LED selectively operable to indicate one or more operating conditions of the container **100** (e.g., green if normal operation, red if abnormal operation, such as low battery charge or inadequate cooling for sensed ambient temperature, etc.). Though the visual display is described in connection with the container system **100**, one of skill in the art will recognize that it can also be applied to all the other implementations discussed herein for the container system **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**.

[0068] In operation, the cooling system **200** can optionally be actuated by pressing a power button. Optionally, the cooling system **200** can additionally (or alternatively) be actuated remotely (e.g., wirelessly) via a remote electronic device **600**, such as a mobile phone, tablet computer, laptop computer, etc. that wirelessly communicates with the cooling system **200** (e.g., with a receiver or transceiver of the circuitry **278**). In still another implementation, the cooling system **200** can automatically cool the chamber **126**, **126'** when the lid **L** is coupled to the vessel **120**, **120'** (e.g., upon receipt by the circuitry, for example in or on the PCBA **278**, of a signal, such as from a pressure sensor, proximity sensor, load sensor, light sensor) that the lid **L** has been coupled with the vessel **120**, **120'**. The chamber **126**, **126'** can be cooled to a predetermined and/or a user selected temperature or temperature range, or automatically cooled to a temperature preset corresponding to the contents in the containers **150** (e.g., insulin, epinephrine, vaccines, etc.). The user selected temperature or temperature range can be selected via a user interface on the container **100** and/or via the remote electronic device **600**.

[0069] The circuitry **278** optionally operates the one or more TECs **220** so that the side of the one or more TECs **220** adjacent the first heat sink **210** is cooled to thereby cool the one or more containers **150** in thermal communication with (e.g., coupled to) the first heat sink **210** and so that the side of the one or more TECs **220** adjacent the one or more second heat sinks **230** is heated. The TECs **220** thereby cool the first heat sink **210** and thereby cools the containers **150** and/or the chamber **126**, **126'**. The container **100** can include one or more sensors (e.g., temperature sensors) **155** operable to sense a temperature of the chamber **126**, **126'**. As best shown in FIG. 7, the one or more sensors **155** can include a temperature sensor that extends through one or more of the prongs of the first heat sink **210** and protrudes from the first heat sink **210** into the chamber **126**, **126'** when the lid **L** is coupled to the vessel **120**, **120'**. The one or more sensors **155** can communicate information to the circuitry **278** indicative of the sensed temperature(s) via the one or more electrical contacts **281**, **282** when the lid **L** is coupled to the vessel **120**, **120'**. The circuitry (e.g., in or on the PCBA **278**) operates one or more of the TECs **220** and one or more fans **280** based at least in part on the sensed temperature information (from the one or more sensors **155**) to cool the first heat sink **210** and/or the chamber **126**, **126'** to the predetermined temperature (e.g., temperature preset) and/or user selected temperature. The circuitry operates the one or more fans **280** to flow air (e.g., received via the intake vents **203**) over the one or more second heat sinks **230** to dissipate heat therefrom, thereby allowing the one or more second heat sinks **230** to draw more heat from the one or more TECs **220**, which in turn allows the one or more TEC's **220** to draw more heat from (i.e., cool) the first heat sink **210** and optionally the chamber **126**, **126'**. Said air flow, once it passes over the one or more second heat sinks **230**, is exhausted via the exhaust vents **205**.

[0070] With reference to FIG. 2, a power base **300** can receive the container **100** thereon and can provide power to the electronics in the container **100** to, for example, charge the one or more batteries **277** or provide power directly to the TECs **220** and/or fan **280**. In one implementation, the power base **300** has an electrical cord that ends in an electrical connector (wall plug, USB connector), which allows the power base **300** to connect to a power source (e.g., wall outlet, USB

connector of power source, such as a laptop or desktop computer). In one implementation, the power base **300** transmits power to the container **100** via inductive coupling. In another implementation, the power base **300** transmits power to the container **100** via one or more electrical contacts (e.g., electrical contact pads, Pogo pins) that contact one or more electrical contacts (e.g., electrical contact pads, contact rings) on the container **100** (e.g., on the bottom **275** of the container **100**).

[0071] FIG. **6** shows a power base **300'** that can receive the container **100** thereon and can provide power to the electronics in the container **100** to, for example, charge the one or more batteries **277** or provide power directly to the TEC **220** and/or fan **280**. The power base **300'** is similar to the power base **300** except as described below. In one implementation, the power base **300'** has an electrical cord that ends in an electrical connector (for a car charger), which allows the power base **300'** to connect to a car charger. Advantageously, the power base **300'** is sized to fit in a cup holder of an automobile, allowing the container **100** to be placed in the cupholder while on the power base **300'**, keeping the container **100** in a substantially stable upright orientation.

[0072] In one variation, the container system **100** is powered using 12 VDC power (e.g., from one or more batteries **277** or power base **300'**). In another variation, the container system **100** is powered using 120 VAC or 240 VAC power, for example using the power base **300**. The circuitry **278** in the container **100** can include a surge protector to inhibit damage to the electronics in the container **100** from a power surge.

[0073] FIG. **9** shows a block diagram of a communication system for (e.g., incorporated into) the devices described herein (e.g., the one or more container systems **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**). In the illustrated embodiment, circuitry EM (e.g., on the PCBA **278**) can receive sensed information from one or more sensors S1-Sn (e.g., level sensors, volume sensors, temperature sensors, such as sensors **155**, battery charge sensors, biometric sensors, load sensors, Global Positioning System or GPS sensors, radiofrequency identification or RFID reader, etc.). The circuitry EM can be housed in the container, such as in the vessel **120**, **120'**, **120E**, **120F**, **120G**, **120H**, **120I**, **120J**, **120K** (e.g., bottom of vessel **120**, **120'**, **120E**, **120F**, **120G**, **120H**, **120I**, **120J**, **120K**, **120L** side of vessel **120**, **120'**, **120E**, **120F**, **120G**, **120H**, **120I**, **120J**, **120K**, **120L** as discussed above) or in a lid L of the container. The circuitry EM can receive information from and/or transmit information (e.g., instructions) to one or more heating or cooling elements HC, such as the TEC **220**, **220E**, **220F1**, **220F2**, **220G**, **220L** (e.g., to operate each of the heating or cooling elements in a heating mode and/or in a cooling mode, turn off, turn on, vary power output of, etc.) and optionally to one or more power storage devices PS (e.g., batteries **277**, **277E**, **277F**, **277L** such as to charge the batteries or manage the power provided by the batteries to the one or more heating or cooling elements **220**, **220E**, **220F1**, **220F2**, **220G**, **220L**).

[0074] Optionally, the circuitry EM can include a wireless transmitter, receiver and/or transceiver to communicate with, e.g., transmit information, such as sensed temperature, position data, to and receive information, such as user instructions, from one or more of: a) a user interface UI1 on the unit (e.g., on the body of the vessel **120**, **120E**, **120F**, **120G**, **120H**, **120I**, **120J**, **120K**, **120L**), b) an electronic device ED (e.g., a mobile electronic device such as a mobile phone, PDA, tablet computer, laptop computer, electronic watch, a desktop computer, remote server), c) the cloud CL (e.g., a cloud-based data storage system), or d) communicate via a wireless communication system such as WiFi and Bluetooth BT. The electronic device ED (such as electronic device **600**) can have a user interface UI2 (such as GUI **610**), that can display information associated with the operation of the container system, and that can receive information (e.g., instructions) from a user and communicate said information to the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** (e.g., to adjust an operation of the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L**).

[0075] In operation, the container system **100** can operate to maintain one or both of the first heat sink **210** and the chamber **126**, **126'** of the vessel **120**, **120'** at a preselected temperature or a user

selected temperature. The cooling system **200** can operate the one or more TECs **220** to cool the first heat sink **210** and, optionally the chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126L** (e.g., if the temperature of the first heat sink **210** or chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126L** is above the preselected temperature, such as when the ambient temperature is above the preselected temperature) or to heat the first heat sink **210** and, optionally chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126L** (e.g., if the temperature of the first heat sink **210** or chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126L** is below the preselected temperature, such as when the ambient temperature is below the preselected temperature). The preselected temperature may be tailored to the contents of the container (e.g., a specific medication, a specific vaccine, insulin pens, epinephrine pens or cartridges, etc.), and can be stored in a memory of the container **100**, and the cooling system **200** or heating system, depending on how the temperature control system is operated, can operate the TEC **220** to approach the preselected or set point temperature.

[0076] Optionally, the circuitry EM can communicate (e.g., wirelessly) information to a remote location (e.g., cloud based data storage system, remote computer, remote server, mobile electronic device such as a smartphone or tablet computer or laptop or desktop computer) and/or to the individual carrying the container (e.g., via their mobile phone, via a visual interface on the container, etc.), such as a temperature history of the first heat sink **210**, **210E1**, **210E2**, **210F1**, **210F2**, **210L** and/or chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126L** to provide a record that can be used to evaluate the efficacy of the medication in the container and/or alerts on the status of the medication in the container **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**. Optionally, the temperature control system (e.g., cooling system, heating system) **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L** automatically operates the TEC **220**, **220E**, **220F1**, **220F2**, **220L** to heat or cool the first heat sink **210**, **210E1**, **210E2**, **210F1**, **210F2**, **210L** and, optionally, the chamber **126**, **126'**, **120E**, **120F1**, **120F2** of the vessel **120**, **120'**, **120E**, **120F** to approach the preselected temperature. In one implementation, the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L** can cool and maintain one or both of the chamber **126**, **126'**, **126E**, **126F1**, **126F1**, **126G1**, **126L** and the containers **150** at or below 15 degrees Celsius, such as at or below 10 degrees Celsius, in some examples at approximately 5 degrees Celsius.

[0077] In one implementation, the one or more sensors **S1-Sn** can include one more air flow sensors in the lid **L** that can monitor airflow through one or both of the intake vent **203** and exhaust vent **205**. If said one or more flow sensors senses that the intake vent **203** is becoming clogged (e.g., with dust) due to a decrease in air flow, the circuitry EM (e.g., on the PCBA **278**) can optionally reverse the operation of the fan **280**, **280E**, **280F** for one or more predetermined periods of time to draw air through the exhaust vent **205** and exhaust air through the intake vent **203** to clear (e.g., unclog, remove the dust from) the intake vent **203**. In another implementation, the circuitry EM can additionally or alternatively send an alert to the user (e.g., via a user interface on the container **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**, wirelessly to a remote electronic device such as the user's mobile phone via GUI **610**) to inform the user of the potential clogging of the intake vent **203**, so that the user can inspect the container **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** and can instruct the circuitry EM (e.g., via an app on the user's mobile phone) to run a "cleaning" operation, for example, by running the fan **280**, **280E**, **280F** in reverse to exhaust air through the intake vent **203**.

[0078] In one implementation, the one or more sensors **S1-Sn** can include one more Global Positioning System (GPS) sensors for tracking the location of the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**. The location information can be communicated, as discussed above, by a transmitter and/or transceiver associated with the circuitry EM to a remote location (e.g., a mobile electronic device, a cloud-based data storage system, etc.).

[0079] In another variation, the circuitry **278** and one or more batteries **277** can be in a removable pack (e.g., DeWalt battery pack) that attaches to the distal end **124** of the vessel **120**, **120'**, **120E**, **120F**, where one or more contacts in the removable pack contact one or more contacts on the distal

end **124** of the vessel **120**, **120'**, **120E**, **120F**, **120G**. The one or more contacts on the distal end **124** of the vessel **120**, **120'**, **120E**, **120F**, **120G** are electrically connected (via one or more wires or one or more intermediate components) with the electrical connections on the proximal end **122** of the vessel **120**, **120E**, **120F**, **120G**, **120H**, **120I**, **120J**, **120K**, or via as discussed above, to provide power to the components of the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L**.

[0080] FIGS. **10A-10B** show a container system **100E** (e.g., capsule container) that includes a cooling system **200E**. The container system **100E** and cooling system **200E** are similar to the container system **100** and cooling system **200** described above in connection with FIGS. **1-8**. Thus, references numerals used to designate the various components of the container vessel **100E** and cooling system **200E** are identical to those used for identifying the corresponding components of the container system **100** and cooling system **200** in FIGS. **1-8**, except that an “E” is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100** and cooling system **200** in FIGS. **1-8** is understood to also apply to the corresponding components of the container system **100E** and cooling system **200E** in FIGS. **10A-10B**, except as described below.

[0081] The container system **100E** differs from the container system **100** in that the opening **123E** in the vessel **120E** has an oval shape and the open chamber **126E** has an oval cross-section. The chamber **126E** is sized to receive a pair of containers **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) side-by-side therein. The container **100E** has electrical contacts **282E** that can interface with electrical contacts **281E** in the lid **L**.

[0082] The lid **L** can have a pair of spaced apart plates **211E1**, **211E2** that can hold the pair of containers (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) therebetween, such as in slots between the plates **211E1**, **211E2**. The plates **211E1**, **211E2** can be part of the first heat sink **210E** in thermal communication with one or more TECs **220E**, such as Peltier element(s), and be in thermal communication with the chamber **126E** of the vessel **120E** (when the lid **L** is attached to the vessel **120E**). As shown in FIG. **10B**, the plates **211E1**, **211E2** can be interposed between the containers **150** (medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) and the inner wall **126AE** of the chamber **126E**.

[0083] The chamber **126E** can be approximately $\frac{1}{2}$ as large as the chamber **126** of vessel **120** (which is sized to hold up to four containers **150**). The other half of the vessel **100E** can house one or more batteries **277E** therein. The chamber **126E** can be insulated (e.g., vacuum insulated) relative to the outer wall **121E** of the vessel **120E**.

[0084] FIGS. **11A-11C** show a container system **100F** (e.g., capsule container) that includes a cooling system **200F**. The container system **100F** and cooling system **200F** are similar to the container system **100** and cooling system **200** described above in connection with FIGS. **1-8**. Thus, references numerals used to designate the various components of the container system **100F** and cooling system **200F** are identical to those used for identifying the corresponding components of the container system **100** and cooling system **200** in FIGS. **1-8**, except that an “F” is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100** and cooling system **200** in FIGS. **1-8** is understood to also apply to the corresponding components of the container vessel **100F** and cooling system **200F** in FIGS. **11A-11C**, except as described below.

[0085] The container system **100F** differs from the container system **100** in that the vessel **120F** has two openings **123F1**, **123F2** at the top of two separate and spaced apart chambers **126F1**, **126F2**. Optionally, the openings **123F1**, **123F2** has a circular shape and each of the chambers **126F1**, **126F2** has a circular cross-section. Each of the chambers **126F1**, **126F2** is sized to receive a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) side-by-side therein. The container vessel **100F** has two separate groups of

electrical contacts **282F1**, **282F2** that can interface with electrical contacts **281F1**, **281F2** in the lid L.

[0086] The lid L can have a pair of spaced apart heat sinks **210F1**, **210F2**, each sized to resiliently hold one container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.), for example in a slot defined by the heat sinks **210F1**, **210F2**. Each of the heat sinks **210F1**, **210F2** can be in thermal communication with a separate TEC **220F1**, **220F2**, which in turn can optionally be in thermal communication with separate second heat sinks (not shown) in the lid L. As discussed in FIGS. 1-8, the cooling system **200F** can have one or more fans **280F** operable to draw air over the second heat sinks (not shown) in the lid L. The chambers **126F1**, **126F2** can be insulated (e.g., vacuum insulated) relative to each other and relative to the outer wall **121F** of the vessel **100F**.

[0087] Advantageously, the heat sinks **210F1**, **210F2** can be operated independently of each other. Accordingly, in one implementation both heat sinks **210F1**, **210F2** are operable to cool the containers **150** to the approximately the same temperature (e.g., down to approximately 5 degrees Celsius) when the containers **150** are in the chambers **126F1**, **126F2** and the lid L is disposed on top of the vessel **120F** to seal the vessel **120F**. In another implementation both heat sinks **210F1**, **210F2** are operable to cool the containers **150** to different temperatures when the containers **150** are in the chambers **126F1**, **126F2** and the lid L is disposed on top of the vessel **120F** to seal the vessel **120F**. In another implementation, for example when a user is ready or almost ready to consume the medicine in the container **100F**, one of the heat sinks **210F1** can be heated to heat its associated container **150** (e.g., to a predetermined consumption or administration temperature, for example to body temperature, to room temperature), while the other heat sink **210F2** cools its associated container **150** in the associated chamber **126F2**. In still another implementation, both heat sinks **210F1**, **210F2** are operated to heat their associated containers **150** (e.g., to the same temperature, to different temperatures).

[0088] FIGS. 12A-12C show a container system **100G** (e.g., a capsule container) that includes a cooling system **200G**. The container system **100G** and cooling system **200G** are similar to the container system **100F** and cooling system **200F** described above in connection with FIGS. 11A-11C. Thus, references numerals used to designate the various components of the container system **100G** and cooling system **200G** are identical to those used for identifying the corresponding components of the container system **100F** and cooling system **200F** in FIGS. 11A-11C, except that a "G" instead of an "F" is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100F** and cooling system **200F** in FIGS. 11A-11C is understood to also apply to the corresponding components of the container vessel **100G** and cooling system **200G** in FIGS. 12A-12C, except as described below. For clarity, FIG. 12A only shows one chamber **126G1**, but can have two chambers **126G1**, **126G2** similar to chambers **126F1**, **126F2** described above. Optionally, the chamber(s) **126G1**, **126G2** are removable from the container system **100G**, as further described below.

[0089] The container system **100G** differs from the container system **100F** in that the heat sink **210G1** is a removable sleeve **210G1** that removably couples to the container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.). The sleeve **210G1** can be made of a thermally conductive material (e.g., a metal, such as aluminum). The sleeve **210G1** can be removed along with the container **150** from the container vessel **120G** (e.g., for placement in a user's purse, backpack, work bag during a commute or travel, etc.). Optionally, the sleeve **210G1** can maintain the container **150** in a cooled state for an extended period of time (e.g., between about 1 hour and about 10 hours, between about 1 hour and about 5 hours, between about 1 hour and about 3 hours, about 2 hours, etc.). When the sleeve **210G1** is coupled with the container **150** and inserted into the chamber **126G1**, the sleeve **210G1** can interface with the cooling system **200G** and operate as a heat transfer interface between the cooling system **200G** (e.g., between one or more TECs **220G** of the cooling system **200G** and the container **150**) to help

cool and/or heat the container **150**. For example, when the cooling system **200G** is used to cool the container **150**, the sleeve **210G1** can function as a heat sink to remove heat (e.g., cool) the container **150** that is attached to the sleeve **210G1**.

[0090] With reference to FIG. **12C**, the sleeve **210G1** can have a top surface **210G2**, an outer wall **210G3** and an inner wall **210G4**, where at least a portion of the inner wall **210G4** can be in contact with the container **150** when the sleeve **210G1** is coupled to the container **150**. Optionally, the sleeve **210G1** can define a cavity (e.g., an annular cavity) **210G5** between the outer wall **210G3** and the inner wall **210G4**. In one implementation, the cavity **210G5** can house a thermal mass material **130G**. In one implementation, the thermal mass material **130G** is a phase change material PCM (e.g., a solid-solid PCM, a solid-fluid PCM) that can transition from a heat absorbing state to a heat releasing state at a transition temperature. In another implementation, the cavity **210G5** is excluded and the sleeve **210G1** instead has a wall that extends between the inner surface **210G4** and the outer wall **210G3** with a thermal surface that can absorb and release heat.

[0091] The sleeve **210G1** can optionally include a heater **210G6** (e.g., a flex heater) in thermal communication with the inner wall **210G4** (e.g., the heater **210G6** can be disposed on the inner wall **210G4**, embedded in the inner wall **210G4**, disposed behind the inner wall **210G4** (e.g., disposed in the cavity **210G5**)). The sleeve **210G1** can have one or more electrical contacts **210G7** on a surface thereof (e.g., on the top surface **210G2**). The one or more electrical contacts **210G7** can be in electrical communication with the heater **210G6**. In another implementation, the sleeve **210G1** can exclude the heater **210G6** and one or more electrical contacts **210G7**.

[0092] In operation, while the sleeve **210G1** is coupled to the container **150** and inserted into the container vessel **120G** with the lid **L** in the closed position relative to the container vessel **120G**, the cooling system **200G** can operate to cool one or both of the chamber **126G1** and the sleeve **210G1**. For example, one or more TECs **220G** of the cooling system **200G** can cool a heat sink surface that contacts the top surface **210G2** of the sleeve **210G1**, thereby also being placed in thermal communication with the inner wall **210G4**, outer wall **210G3** and optional thermal mass **130G** (e.g., PCM) in the cavity **210G5**. The TECs **220G** can thereby cool the sleeve **210G1** and thereby cool the container **150** attached to it, as well as charge the optional thermal mass **130G** (e.g., PCM). Optionally, where the sleeve **210G1** includes the heater **210G6**, a controller of the system **200G** can operate the heater **210G6** to heat the contents of the container **150** (e.g., to room temperature, body temperature) prior to the container **150** being removed from the container vessel **120G** for use (e.g. for application of the contents of the container to the user, such as via an injector pen). For example, the controller can provide power to the heater **210G6** via the electrical contacts **210G7** that contact electrical contacts in the lid **L** when the lid **L** is in a closed position relative to the container vessel **100**.

[0093] In one implementation, once the cooling system **200G** has cooled the sleeve **210G1** and its attached container **150**, the user can optionally remove the sleeve **210G1** with its attached container **150** from the container vessel **120G**, as described above (e.g., for travel, commute, etc.) and the charged thermal mass **130G** can maintain the container **150** attached to the sleeve **210G1** in a cooled state for an extended period of time, as discussed above.

[0094] FIG. **13** shows another implementation of a chamber **126G1** in the container system **100G** (e.g., a capsule container) that includes a cooling system **200G**. As discussed above, the chamber **126G1** can receive a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) attached to the sleeve **210G1**. The chamber **126G1** can be actuated between a retracted position and an extended position in the container vessel **100G**. As shown in FIG. **13**, the chamber **126G1** can be spring loaded within the container vessel **100G**. A guide **430** can guide the movement of the chamber **126G1** between the retracted and extended position.

[0095] In one implementation, the chamber **126G1** can have an actuation mechanism **400** that can optionally include a spring **410** that extends between a bottom of the chamber **126G1** and a cam

420. The spring **410** can be a compression spring. In one implementation, the cam **240** can move between a first orientation to position the chamber **126G1** in the retracted position and a second orientation to position the chamber **126G2** in the extended position. The movement of the cam **240** to change its orientation can be actuated by pushing down on the sleeve **210G1** (e.g., on the top surface **210G2** of the sleeve **210G1**). Movement of the chamber **126G1** to the extended position can facilitate removal of the container **150** (e.g., with the attached sleeve **210G1**) from the chamber **126G1** (e.g., when ready for use by the user, as discussed above).

[0096] Optionally, with the chamber **126G1** in the extended position, and with the container **150** in the chamber **126G1** and attached to the sleeve **210G1**, movement of the lid **L** to the closed position relative to the container vessel **120G** can urge the chamber **126G1** into the container vessel **120G** and actuate the movement of the cam **420** to allow the chamber **126G1** to move to the retracted position. Though the actuation mechanism **400** is described in connection with the chamber **126G1** and container system **100G**, one of skill in the art will recognize that the features of the actuation mechanism **400** described herein can also be applied to all the other implementations discussed herein for the container system **100**, **100E**, **100F**, **100G**.

[0097] FIGS. **14A-14B** shows another implementation of a chamber **126G1** in the container system **100G** (e.g., a capsule container) that includes a cooling system **200G**. As discussed above, the chamber **126G1** can receive a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) attached to the sleeve **210G1**. The chamber **126G1** can be actuated between a retracted position and an extended position in the container vessel **120G**. As shown in FIGS. **14A-14B**, the chamber **126G1** can be actuated between the retracted position and the extended position by an actuation mechanism **400'**. The actuation mechanism **400'** can optionally be housed in the container vessel **120G** below the chamber **126G1** (e.g., between a bottom of the chamber **126G1** and a bottom of the container vessel **120G**). A guide **430** can guide the movement of the chamber **126G1** between the retracted and extended position.

[0098] With reference to FIG. **14B**, the actuation mechanism **400'** can include a linear actuator **410'** and a motor **420'** operable to drive the linear actuator **410'**. The linear actuator **410'** can optionally include a coupling that couples to an output shaft of the motor **420'**. The coupling **412'** is coupled to a ball screw **414'** that rotates when the motor **420'** rotates the coupling **412'**. The ball screw **414'** rotates relative to a ball screw nut **416'**, where the ball screw nut **416'** travels along the ball screw **414'** as the motor **420'** rotates the coupling **412'** (e.g., travels rightward in the drawing when coupling **412'** rotates in one direction and travels leftward in the drawing when the coupling **412'** rotates in the opposite direction). The ball screw nut **416'** can be attached to a rod such that the rod translates (at least partially within a bushing **419'**) along the axis of the ball screw **414'** as the screw **414'** rotates. An end of the rod **418'** can engage a bottom of the chamber **126G1** to move the chamber **126G1** between the retracted and extended position relative to the container vessel **120G**, however, in other implementations, the actuation mechanism **400'** can be other suitable linear motion mechanisms (e.g., instead of an electric motor **420'** can include a pneumatic or hydraulic system to translate the rod **418'**). Though the actuation mechanism **400'** is described in connection with the chamber **126G1** and container vessel **120G**, one of skill in the art will recognize that the features of the actuation mechanism **400'** described herein can also be applied to all the other implementations discussed herein for the container vessel **100**, **100E**, **100F**, **100G**.

[0099] FIG. **15** shows another implementation of a chamber **126G1** in the container system **100G** (e.g., a capsule container) that includes a cooling system **200G**. As discussed above, the chamber **126G1** can receive a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) attached to the sleeve **210G1**. The chamber **126G1** can be actuated between a retracted position and an extended position in the container vessel **120G**. As shown in FIG. **15**, the chamber **126G1** can be actuated between the retracted position and the extended position by an actuation mechanism **400''**. The actuation mechanism **400''** can optionally be housed in the lid **L**. Though not shown, a guide (similar to guide **430**) can guide the movement

of the chamber **126G1** between the retracted and extended position.

[0100] With reference to FIG. **15**, the actuation mechanism **400''** can include a magnet **420''**. In one implementation, the magnet **420''** can be an electromagnet. In operation, the electromagnet **420''** can be operated to draw the sleeve **210G1** (e.g., the top surface **210G2** of the sleeve **210G1**) into contact with a heat sink surface and/or one or more TECs **220G** to place the sleeve **210G1** (and therefore the container **150** coupled to the sleeve **210G1**) in thermal communication with the one or more TECs **220G**, which can be operated to cool the sleeve **210G1** and/or container **150** and/or the chamber **126G1**. The electromagnet **420''** can be turned off or not operated to allow the sleeve **210G1** (and container **150** attached to it) to be displaced from the heat sink and/or one or more TECs **220G** to thereby thermally disconnect the container **150** and sleeve **210G1** from the TECs **220G**. The electromagnet **420''** can be turned off or disengaged when, for example, the user wishes to remove the container **150** and sleeve **210G1** from the container vessel **120G** (e.g., for storing in another compartment, such as a purse, backpack, travel bag, etc. during a commute or trip). Though the actuation mechanism **400''** is described in connection with the chamber **126G1** and container vessel **120G**, one of skill in the art will recognize that the features of the actuation mechanism **400''** described herein can also be applied to all the other implementations discussed herein for the container vessel **100**, **100E**, **100F**, **100G**.

[0101] In another implementation, the container system **100**, **100E**, **100F**, **100G** can have chambers **126**, **126E**, **126F1**, **126F2**, **126G1** that can be completely removed from the container vessel **120**, **120E**, **120F**, **120F**, such as for travel or commute, where the chamber can hold the container **150** (e.g., vial, cartridge (such as for use in injector pen), injector pen, etc.) therein (e.g., provide a travel pack) until the container **150** is ready for use.

[0102] FIGS. **16A-16C** show a container system **100H** (e.g., a capsule container) that includes a cooling system **200H**. The container system **100H** and cooling system **200H** are similar to the container system **100G** and cooling system **200G** described above in connection with FIGS. **12A-12C**. Thus, references numerals used to designate the various components of the container system **100H** and cooling system **200H** are identical to those used for identifying the corresponding components of the container system **100G** and cooling system **200G** in FIGS. **12A-12C**, except that an “H” instead of a “G” is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100G** and cooling system **200G** in FIGS. **12A-12C** is understood to also apply to the corresponding components of the container system **100H** and cooling system **200H** in FIGS. **16A-16C**, except as described below.

[0103] As shown in FIG. **16**, the container system **100H** has a container vessel **120H** and a lid **L**. The lid **L** can include a cooling system **200G**. The container vessel **120H** can optionally have one or more chambers **126H** that extend to corresponding one or more openings **123H**. Though FIG. **16** shows the container vessel **120H** having six chambers **126H**, one of skill in the art will recognize that the container vessel **120H** can have more or fewer chambers **126H** than shown in FIG. **16**. The chamber(s) **126H** of the container vessel **120H** can removably hold a corresponding capsule **210H** therein. In one implementation, the container vessel **120H** can have the same or similar structure as shown and described above for the container vessel **120**, **120E**, **120F**, **120G**. Optionally, the container vessel **120H** can have a cavity between the chamber(s) **126H** and the outer surface of the container vessel **120H** that is vacuum insulated. In another implementation, the container vessel **120H** excludes vacuum insulation and can instead have a gap or cavity between the chamber(s) **126H** and an outer surface of the container vessel **120H** that is filled with air. In still another implementation, the container vessel **120H** can have a gap or cavity between the chamber(s) **126H** and an outer surface of the container vessel **120H** that includes an insulating material.

[0104] With continued reference to FIG. **16**, the capsule(s) **210H** have a vessel portion **210H1** and a lid portion **210H2** that together can enclose a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.). The lid portion **210H2** can be moved between a closed position relative to (e.g., adjacent) the vessel portion **210H1** and an open

position relative to (e.g., spaced apart from) the vessel portion **210H1**. In the closed position, the lid portion **210H2** can optionally be held against the vessel portion **210H1** (e.g., by one or more magnetic surfaces of the lid portion **210H2** and/or vessel portion **210H1**) to inhibit (e.g., prevent) the container **150** from inadvertently falling out of the capsule **210H**.

[0105] FIG. **16A** shows one implementation of a capsule **210H**, where the vessel portion **210H1** and lid portion **210H2** have an outer surface **210H3** and an inner surface **210H4** that defines a cavity **210H8** that receives the container **150**. The vessel portion **210H1** and lid portion **210H2** can also have one or more intermediate walls **210H6** radially between the inner surface **210H4** and the outer surface **210H3** that define a first cavity **210H5** between the inner wall **210H4** and the intermediate wall(s) **210H6** and a second cavity **210H9** between the intermediate wall(s) **210H6** and the outer surface **210H3**. Optionally, the second cavity **210H5** can be vacuum insulated (i.e., the second cavity **210H5** can be under vacuum or negative pressure force). Optionally, the first cavity **210H5** can house a thermal mass material **130H**. In one implementation, the thermal mass material **130H** is a phase change material PCM (e.g., a solid-solid PCM, a solid-fluid PCM) that can transition from a heat absorbing state to a heat releasing state at a transition temperature. In another implementation, the cavity **210H5** is excluded and the capsule **210H** instead has a wall that extends between the inner surface **210H4** and the intermediate wall(s) **210H6** that can absorb and release heat.

[0106] With continued reference to FIG. **16A**, the capsule **210H** has a thermally conductive contact **210H7** at one or both ends of the capsule **210H**. The thermally conductive contact **210H7** can be made of metal, though it can be made of other thermally conductive material. In one implementation, the thermally conductive contact **210H7** is made of copper. The thermally conductive contact **210H** can extend from the outer surface **210H3** to the inner surface **210H4** and through the first and second cavities **210H5**, **210H9**, so as to be in thermal contact with the thermal mass material **130H**.

[0107] In operation, when a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) is inserted into the capsule **210H** (e.g. into the vessel portion **210H1** and lid portion **210H2**) and then inserted into the chamber **126H**, and the lid **L** closed over the container vessel **120H**, the thermally conductive contact(s) **210H7** will be placed in thermal communication (e.g., thermally contact, directly contact) with a cold-side heat sink of the cooling system **200G** (e.g., similar to the heat sink **210** in FIG. **4**) that is itself in thermal communication with one or more TECs (e.g., similar to TEC **220** in FIG. **4**), where the one or more TECs are operated to remove heat from (e.g., cool) the cold side heat sink, which in turn removes heat from (e.g., cools) the thermally conductive contact(s) **210H7**. The thermally conductive contact(s) **210H7** in turn remove heat from the cavity **210H8** to thereby cool the container **150**, as well as remove heat from the thermal mass material **130H** in the cavity **210H5** to thereby charge the thermal mass material **130H**. In one implementation, the cold side heat sink thermally contacts one of the thermally conductive contacts **210H7**. In another implementation, the cold side heat sink thermally contacts both of the thermally conductive contacts **210H7**. For example, the cold side heat sink in the lid **L** can thermally contact the thermally conductive contact **210H7** at one end of the capsule **210H** as well as thermally contact an inner wall of the chamber **126H** that itself contacts the thermally conductive contact **210H7** at the opposite end of the capsule **210H**.

[0108] The capsule **210H** can be removed along with the container **150** (e.g., one at a time, two at a time, etc.) from the container vessel **120H** (e.g., for placement in a user's purse, backpack, work bag during a commute or travel, etc.). Optionally, the capsule **210H** can maintain the container **150** in a cooled state for an extended period of time (e.g., between about 1 hour and about 15 hours, about 14 hours, between about 1 hour and about 10 hours, between about 1 hour and about 3 hours, about 2 hours, etc.). The capsule **210H** can maintain the container **150** approximately at a temperature of about 2-8 degrees Celsius. When the capsule **210H** receives or houses the container **150** and is then inserted into the chamber **126H** of the container vessel **120H**, the capsule **210H** can

interface with the cooling system **200H** and operate as a heat transfer interface between the cooling system **200H** (e.g., between one or more TECs **220H** of the cooling system **200H** and the container **150**) to help cool and/or heat the container **150**. For example, when the cooling system **200H** is used to cool the container **150**, the capsule **210H** can function as a heat sink to remove heat (e.g., cool) the container **150** that is disposed in the capsule **210H**.

[0109] In one implementation, the cooling system **200H** receives power via a power cord PC that can be connected to a wall outlet. However, the power cord PC can have other suitable connectors that allow the cooling system **200H** to receive power from a power source other than a wall outlet. Power can be provided from the container vessel **120H**, to which the power cord PC is connected, to the cooling system **200H** in the lid via one or more electrical contacts on a rim of the container vessel **120H** and on the lid L (e.g., similar to electrical contacts **282** described above in connection with FIG. 3). In another implementation, the power cord PC is excluded and the container vessel **120H** can have one or more batteries (such as batteries **277** in FIG. 4) that provide power to the cooling system **200H** (e.g., via electrical contacts, such as contacts **282** in FIG. 3) when the lid L is disposed over the container vessel **120H**.

[0110] FIGS. **16B-16C** show another implementation of the capsule **210H'** for use with a container system **100H'** and cooling system **200H'**. The capsule **210H'**, container system **100H'** and cooling system **200H'** are similar to the capsule **210H**, container system **100H** and cooling system **200H** described above in connection with FIGS. **16-16A**. Thus, references numerals used to designate the various components of the capsule **210H**, container system **100H** and cooling system **200H** are identical to those used for identifying the corresponding components of the capsule **210H'**, container system **100H'** and cooling system **200H'** in FIGS. **16B-16C**, except that an "'" is added to the numerical identifier. Therefore, the structure and description for the various components of the capsule **210H**, container system **100H** and cooling system **200H** in FIGS. **16-16A** is understood to also apply to the corresponding components of the capsule **210H'**, container system **100H'** and cooling system **200H'** in FIGS. **16B-16C**, except as described below.

[0111] The capsule **210H'** differs from the capsule **210H** in that the thermally conductive contact(s) **210H7** are excluded. The capsule **210H'** has a movable mass **162H** disposed in the cavity **210H9'** between the intermediate wall **210H6'** and the outer wall **210H3'**. The movable mass **162H** can optionally be a magnet. In another implementation, the movable mass **162H** can be a metal block. The movable mass **162H** can optionally be movably coupled to the intermediate wall **210H6'** by a flexible thermally conductive element **164H**, which operates as a thermal bridge between the movable mass **162H** and the thermal mass material **130H'**. In one implementation, the flexible thermally conductive element **164H** can be made of copper. However, the flexible thermally conductive element **164H** can be made of other suitable thermally conductive materials. The flexible thermally conductive element **164H** can be a leaf spring or similar resilient member that is attached at one end to the intermediate wall **210H6'** and at its other end to the movable mass **162H**. The movable mass **162H** can optionally move within the second cavity **210H9'** (e.g., a vacuum insulated cavity) between a first position where it is in contact with the intermediate wall **210H6'** and a second position where it is in contact with the outer wall **210H3'** of the capsule **210H'**.

[0112] The container vessel **120H'** can include one or more magnets **160H** adjacent a wall of the chamber(s) **126H'**. In one implementation, the one or more magnets **160H** are permanent magnets. In another implementation, the one or more magnets **160H** are electromagnets. The one or more magnets **160H** can be in thermal communication with a cold side heat sink of the cooling system **200H'** (e.g., via a wall or surface of the container vessel **120H'**, such as a wall of the chamber(s) **126H'** that interfaces with the cold side heat sink when the lid L is placed on the container vessel **120H'**).

[0113] In operation, when a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) is inserted into the capsule **210H'** (e.g. into the vessel portion **210H1'** and lid portion **210H2'**) and then inserted into the chamber **126H'**, and the lid L

closed over the container vessel **120H'**, the one or more magnets **160H** in the container vessel **120H'** draw the movable mass **162H** into contact with the outer wall **210H3'** of the capsule **210H'**. The cooling system **200H'** draws heat out of the cavity **210H8'** of the capsule **210H'** (e.g., via operation of one or more TECs to draw heat from cold side heat sink, which itself draws heat from surfaces of components in the container vessel **120H'** in thermal communication with the magnet **160H**) by drawing heat from the thermal mass material **130H'** via the flexible thermally conductive element **164H** and contact between the movable mass **162H**, outer wall **210H3'** and magnet **160H**. As heat is drawn from the thermal mass material **130H'** to charge it, it also draws heat from the cavity **210H8'** via the inner wall **210H4'**. The magnet **160H** and movable mass **162H** (e.g., magnet, metallic component) therefore operate to form a thermal bridge through the cavity **210H9'** (e.g., vacuum insulated cavity) to the thermal mass material **130H'**.

[0114] The capsule **210H'** can be removed along with the container **150** (e.g., one at a time, two at a time, etc.) from the container vessel **120H'** (e.g., for placement in a user's purse, backpack, work bag during a commute or travel, etc.). Optionally, the capsule **210H'** can maintain the container **150** in a cooled state for an extended period of time (e.g., between about 1 hour and about 15 hours, about 14 hours, between about 1 hour and about 10 hours, between about 1 hour and about 3 hours, about 2 hours, etc.). The capsule **210H'** can maintain the container **150** approximately at a temperature of about 2-8 degrees Celsius.

[0115] The capsule(s) **210H**, **210H'** can optionally have a wireless transmitter and/or transceiver and a power source (e.g., battery) disposed therein (e.g., disposed in the cavity **210H9**, **210H9'**), and can have a temperature sensors in communication with the cavity **210H8**, **210H8'** (e.g., in thermal contact with the inner wall **210H4**, **210H4'**). The wireless transmitter and/or transceiver can optionally allow connectivity of the capsule(s) **210H**, **210H'** with an electronic device (e.g., a mobile electronic device, such as a smartphone), such as via an app on the electronic device, and can transmit sensed temperature information to the electronic device for tracking of internal temperature of the capsule **210H'**, **210H**. Optionally, the transmitter and/or transceiver can transmit an alert signal to the electronic device (e.g., visual alert, audible alert), such as a notification via the app, if the sensed temperature exceeds a temperature range (e.g., predetermined temperature range, preselected temperature limit) for the medication in the container **150**. When the capsule **210H**, **210H'** is inserted into the chamber **126H**, **126H'** of the container vessel **120H**, **120H'**, the transmitter and/or transceiver can also wirelessly transmit sensed temperature data sensed by the temperature sensor to the electronic device. Optionally, when in the container vessel **120H**, **120H'**, the battery in the capsule(s) **210H**, **210H'** can be recharged (e.g., via induction power transfer, or via electrical contacts). In addition to maintaining the container **150** (and medication in the container **150**) at or below a predetermined temperature range (e.g., 2-8 degrees C.) for a prolonged period of time (e.g., up to 14 hours, up to 10 hours, up to 5 hours, up to 3 hours, etc.), the capsule(s) **210H**, **210H'** can protect the container **150** therein from damage (e.g., breaking, spillage) if the capsule **210H**, **210H'** is dropped.

[0116] FIGS. **17-17B** show a container system **100I** (e.g., a capsule container) that includes a cooling system **200I**. The container system **100I** and cooling system **200I** are similar to the container system **100H** and cooling system **200H** described above in connection with FIGS. **16-16A**. Thus, references numerals used to designate the various components of the container system **100I** and cooling system **200I** are identical to those used for identifying the corresponding components of the container system **100H** and cooling system **200H** in FIGS. **16-16A**, except that an "I" instead of an "H" is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100H** and cooling system **200H** in FIGS. **16-16A** is understood to also apply to the corresponding components of the container system **100I** and cooling system **200I** in FIGS. **17-17B**, except as described below.

[0117] As shown in FIG. **17**, the container system **100I** has a container vessel **120I** and a lid **L**. The lid **L** can include a cooling system **200I**. The container vessel **120I** can optionally have one or more

chambers **126I** that extend to corresponding one or more openings **123I**, each chamber **126I** sized to receive and hold a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.). Though FIG. **17** shows the container vessel **120I** having six chambers **126I**, one of skill in the art will recognize that the container vessel **120I** can have more or fewer chambers **126I** than shown in FIG. **17**. Optionally, the container vessel **120I** can have a chamber **12612** that extends to an opening **12312**, the chamber **12612** sized to receive a capsule **210I**, which itself can hold one or more (e.g., one, two, etc.) containers **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.), as further described below.

[0118] In one implementation, the container vessel **120I** can have the same or similar structure as shown and described above for the container vessel **120**, **120E**, **120F**, **120G**, **120H**. Optionally, the container vessel **120I** can have a cavity between the chamber(s) **126I** and the outer surface of the container vessel **120I** that is vacuum insulated. In another implementation, the container vessel **120I** excludes vacuum insulation and can instead have a gap or cavity between the chamber(s) **126I** and an outer surface of the container vessel **120I** that is filled with air. In still another implementation, the container vessel **120I** can have a gap or cavity between the chamber(s) **126I** and an outer surface of the container vessel **120I** that includes an insulating material.

[0119] FIGS. **17A-17B** shows one implementation of a capsule **210I** having a vessel portion **210I1** and a lid portion **210I2** (attached via a hinge **211I**) that together can enclose one or more containers **150** (e.g., two containers **150** in FIG. **17A**). The hinge **211I** allows the lid portion **210I2** to be moved between a closed position an open position relative to the vessel portion **210I1**. In the closed position, the lid portion **210I2** can optionally be held against the vessel portion **210I1** (e.g., by one or more magnetic surfaces of the lid portion **210I2** and/or vessel portion **210I1**) to inhibit (e.g., prevent) the container **150** from inadvertently falling out of the capsule **210I**.

[0120] The vessel portion **210I1** and lid portion **210I2** have an outer surface **210I3** and an inner surface **210I4** that defines a cavity **210I8** that receives the container(s) **150**. The vessel portion **210I1** and lid portion **210I2** can also have an intermediate wall **210I6** radially between the inner surface **210I4** and the outer surface **210I3** that define a first cavity **210I5** between the inner wall **210I4** and the intermediate wall **210I6** and a second cavity **210I9** between the intermediate wall **210I6** and the outer surface **210I3**. Optionally, the second cavity **210I5** can be vacuum insulated (i.e., the second cavity **210I5** can be under vacuum or negative pressure force). Optionally, the first cavity **210I5** can house a thermal mass material **130I**. In one implementation, the thermal mass material **130I** is a phase change material PCM (e.g., a solid-solid PCM, a solid-fluid PCM) that can transition from a heat absorbing state to a heat releasing state at a transition temperature. In another implementation, the cavity **210I5** is excluded and the capsule **210I** instead has a wall that extends between the inner surface **210I4** and the intermediate wall(s) **210I6** that can absorb and release heat.

[0121] In operation, when a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) is inserted into the capsule **210I** (e.g. into the vessel portion **210I1**) and then inserted into the chamber **126I**, and the lid **L** closed over the container vessel **120I**, the lid portion **210I2** can be in the open position relative to the vessel portion **210I1** (see FIGS. **17**, **17A**), allowing the thermal mass material **130I** in the cavity **210I5** to be placed in thermal communication (e.g., thermally contact, directly contact) with a cold-side heat sink of the cooling system **200I** (e.g., similar to the heat sink **210** in FIG. **4**) that is itself in thermal communication with one or more TECs (e.g., similar to TEC **220** in FIG. **4**), where the one or more TECs are operated to remove heat from (e.g., cool) the cold side heat sink, which in turn removes heat from (e.g., cools) the thermal mass material **130I** and cavity **210I8** in the capsule **210I**, as well as any containers **150** in the capsule **210I**.

[0122] The capsule **210I** can be removed along with one or more containers **150** (e.g., one at a time, two at a time, etc.) from the container vessel **120I** (e.g., for placement in a user's purse,

backpack, work bag during a commute or travel, etc.). Optionally, the capsule **210I** can maintain the container(s) **150** in a cooled state for an extended period of time (e.g., between about 1 hour and about 15 hours, about 14 hours, between about 1 hour and about 10 hours, between about 1 hour and about 3 hours, about 2 hours, etc.). The capsule **210I** can maintain the container **150** approximately at a temperature of about 2-8 degrees Celsius.

[0123] The capsule **210I** can optionally have a wireless transmitter and/or transceiver and a power source (e.g., battery) disposed therein (e.g., disposed in the cavity **210I9**), and can have a temperature sensors in communication with the cavity **210I8** (e.g., in thermal contact with the inner wall **210I4**). The wireless transmitter and/or transceiver can optionally allow connectivity of the capsule **210I** with an electronic device (e.g., a mobile electronic device, such as a smartphone), such as via an app on the electronic device, and can transmit sensed temperature information to the electronic device for tracking of internal temperature of the capsule **210I**. Optionally, the transmitter and/or transceiver can transmit an alert signal to the electronic device (e.g., visual alert, audible alert), such as a notification via the app, if the sensed temperature exceeds a temperature range (e.g., predetermined temperature range, preselected temperature limit) for the medication in the container **150**. When the capsule **210I** is inserted into the chamber **126I** of the container vessel **120I**, the transmitter and/or transceiver can also wirelessly transmit sensed temperature data sensed by the temperature sensor to the electronic device. Optionally, when in the container vessel **120I**, the battery in the capsule(s) **210I** can be recharged (e.g., via induction power transfer, or via electrical contacts). In addition to maintaining the container **150** (and medication in the container **150**) at or below a predetermined temperature range (e.g., 2-8 degrees C.) for a prolonged period of time (e.g., up to 14 hours, up to 10 hours, up to 5 hours, up to 3 hours, etc.), the capsule **210I** can protect the container **150** therein from damage (e.g., breaking, spillage) if the capsule **210I** is dropped.

[0124] In one implementation, the cooling system **200I** receives power via a power cord PC that can be connected to a wall outlet. However, the power cord PC can have other suitable connectors that allow the cooling system **200I** to receive power from a power source other than a wall outlet. Power can be provided from the container vessel **120I**, to which the power cord PC is connected, to the cooling system **200I** in the lid via one or more electrical contacts on a rim of the container vessel **120I** and on the lid L (e.g., similar to electrical contacts **282** described above in connection with FIG. 3). In another implementation, the power cord PC is excluded and the container vessel **120I** can have one or more batteries (such as batteries **277** in FIG. 4) that provide power to the cooling system **200I** (e.g., via electrical contacts, such as contacts **282** in FIG. 3) when the lid L is disposed over the container vessel **120I**.

[0125] FIGS. **18-18B** show a container system **100J** (e.g., a cartridge container) that includes a cooling system **200J**. The container system **100J** and cooling system **200J** are similar to the container system **100H** and cooling system **200H** described above in connection with FIGS. **16-16A**. Thus, references numerals used to designate the various components of the container system **100J** and cooling system **200J** are identical to those used for identifying the corresponding components of the container system **100H** and cooling system **200H** in FIGS. **16-16A**, except that a “J” instead of an “H” is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100H** and cooling system **200H** in FIGS. **16-16A** is understood to also apply to the corresponding components of the container system **100J** and cooling system **200J** in FIGS. **18-18B**, except as described below.

[0126] As shown in FIG. **18**, the container system **100J** has a container vessel **120J** and a lid L. The lid L can include a cooling system **200J**. The container vessel **120J** can optionally have one or more chambers **126J** that extend to corresponding one or more openings **123J**, each chamber **126J** sized to receive and hold a container **150J** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.). In FIG. **16**, the container **150J** is a cartridge that can be separately inserted into an injector device (e.g., injector pen) **170J** (see FIG. **18B**), as discussed

further below. The container vessel **120J** differs from the container vessel **120H** in that the opening(s) **123J** and chamber(s) **126J** are sized to receive container(s) **150J** that are cartridges. Though FIG. **18** shows the container vessel **120J** having six chambers **126J**, each being sized to removably receive a container **150J** (e.g., a cartridge), one of skill in the art will recognize that the container vessel **120J** can have more or fewer chambers **126J** than shown in FIG. **18**.

[0127] In one implementation, the container vessel **120J** can have the same or similar structure as shown and described above for the container vessel **120**, **120E**, **120F**, **120G**, **120H**, **120I** and can maintain the container(s) **150** in a cooled state of approximately at a temperature of about 2-8 degrees Celsius. Optionally, the container vessel **120J** can have a cavity between the chamber(s) **126J** and the outer surface of the container vessel **120J** that is vacuum insulated. In another implementation, the container vessel **120J** excludes vacuum insulation and can instead have a gap or cavity between the chamber(s) **126J** and an outer surface of the container vessel **120J** that is filled with air. In still another implementation, the container vessel **120J** can have a gap or cavity between the chamber(s) **126J** and an outer surface of the container vessel **120J** that includes an insulating material.

[0128] FIG. **18A** shows one implementation of a container **150J** (e.g. a cartridge, an injector pen) that can optionally house a medication (e.g., epinephrine, insulin, a vaccine, etc.). the container **150J** can have a temperature sensor **152J** and a radiofrequency identification (RFID) tag or chip **154J**, with the temperature sensors **152J** being in communication (e.g., electrically connected) with the RFID chip **154J**. The RFID chip **154J** can store temperature data sensed by the temperature sensor **152J**. Advantageously, the temperature sensor **152J** can track the temperature of the container **150J** from when it leaves the distribution center to when it arrives at a person's (consumer's) home, and to when it needs to be administered. The temperature data sensed by the temperature sensor **152J** is stored in the RFID chip **154J**, thereby providing a temperature history of the container **150J** from when it leaves the distribution center to when it arrives at a person's (consumer's) home, and to when it needs to be administered. In one implementation, the container vessel **120J** can have an optional RFID reader that can read the RFID chip **154J** once the container **150J** is inserted into the chamber **126J** of the container vessel **120J** to capture the temperature history stored in the RFID chip **154J**. Optionally, the container system **100J** can inform the user (e.g., via one or both of a graphical user interface on the container vessel **120J** and an app on an electronic device paired with the container system **100J**) that the medication in the container **150J** (e.g., cartridge) can be delivered (e.g., that the temperature history read from the RFID chip **154J** indicates the medication in the container **150J** has been maintained within a predetermined temperature range, so that the medication is deemed effective for delivery).

[0129] FIG. **18B** shows an injection device **170J** (e.g., auto injection device) into which the container **150J** can be inserted prior to use (e.g., prior to application of the auto injection device on the user to deliver a medication in the container **150J**, such as via a needle of the injection device **170J**). When the container **150J** (e.g., cartridge) is removed from the container vessel **120J** and placed into the injection device **170J**, an optional RFID reader in the injection device **170J** can read the RFID chip **154J** and send an alert to the user (via one or both of a graphical user interface on the injection device **170J** and an app on an electronic device paired with the injection device **170J**) that the medication can be delivered (e.g., that the temperature history read from the RFID chip **154J** indicates the medication in the container **150** has been maintained within a predetermined temperature range, so that the medication is deemed effective for delivery).

[0130] In operation, when a container **150J** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) is inserted into the chamber **126J**, and the lid **L** closed over the container vessel **120J**, the container **150J** can optionally be placed in thermal communication (e.g., thermally contact, directly contact) with a cold-side heat sink of the cooling system **200J** (e.g., similar to the heat sink **210** in FIG. **4**) that is itself in thermal communication with one or more TECs (e.g., similar to TEC **220** in FIG. **4**), where the one or more TECs are

operated to remove heat from (e.g., cool) the cold side heat sink, which in turn removes heat from (e.g., cools) the container(s) **150J** in the vessel container **120J**.

[0131] Optionally, the container **150J** can optionally have a wireless transmitter and/or transceiver and a power source (e.g., battery) disposed therein. The wireless transmitter and/or transceiver can optionally allow connectivity of the container **150J** with an electronic device (e.g., a mobile electronic device, such as a smartphone), such as via an app on the electronic device, and can transmit sensed temperature information (from the temperature sensor **152J**) to the electronic device for tracking of internal temperature of the container **150J** (e.g., in addition to or in place of tracking the sensed temperature history of the container **150J** via the RFID chip **154J**). Optionally, the transmitter and/or transceiver can transmit an alert signal to the electronic device (e.g., visual alert, audible alert), such as a notification via the app, if the sensed temperature exceeds a temperature range (e.g., predetermined temperature range, preselected temperature limit) for the medication in the container **150J**. When the container **150J** is inserted into the chamber **126J** of the container vessel **120J**, the transmitter and/or transceiver can also wirelessly transmit sensed temperature data sensed by the temperature sensor **152J** to the electronic device. Optionally, when in the container vessel **120J**, the battery in the container **150J** can be recharged (e.g., via induction power transfer, or via electrical contacts).

[0132] In one implementation, the cooling system **200J** receives power via a power cord PC that can be connected to a wall outlet. However, the power cord PC can have other suitable connectors that allow the cooling system **200J** to receive power from a power source other than a wall outlet. Power can be provided from the container vessel **120J**, to which the power cord PC is connected, to the cooling system **200J** in the lid via one or more electrical contacts on a rim of the container vessel **120J** and on the lid L (e.g., similar to electrical contacts **282** described above in connection with FIG. 3). In another implementation, the power cord PC is excluded and the container vessel **120J** can have one or more batteries (such as batteries **277** in FIG. 4) that provide power to the cooling system **200J** (e.g., via electrical contacts, such as contacts **282** in FIG. 3) when the lid L is disposed over the container vessel **120J**.

[0133] FIG. **19A** shows a container system **100K** (e.g., a medicine cooler container) that includes a cooling system **200K**. Though the container system **100K** has a generally box shape, in other implementations it can have a generally cylindrical or tube shape, similar to the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**. In one implementation, the cooling system **200K** can be in the lid L of the container system **100K** and can be similar to (e.g., have the same or similar components as) the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**. In another implementation, the cooling system can be disposed in a portion of the container vessel **120K** (e.g. a bottom portion of the container vessel **120K**).

[0134] As shown in FIG. **19A**, the container system **100K** can include a display screen **180K**. Though FIG. **19A** shows the display screen **180K** on the lid L, it can alternatively (or additionally) be incorporated into a side surface **122K** of the container vessel **120K**. The display screen **180K** can be an electronic ink or E-ink display (e.g., electrophoretic ink display). In another implementation, the display screen **180K** can be a digital display (e.g., liquid crystal display or LCD, light emitting diode or LED, etc.). Optionally, the display screen **180K** can display a label **182K** (e.g., a shipping label with one or more of an address of sender, an address of recipient, a Maxi Code machine readable symbol, a QR code, a routing code, a barcode, and a tracking number). The container system **100K** can also include a user interface **184K**. In FIG. **19A**, the user interface **184K** is a button on the lid L. In another implementation, the user interface **184K** is disposed on the side surface **122K** of the container vessel **120K**. In one implementation, the user interface **184K** is a depressible button. In another implementation, the user interface **184K** is a capacitive sensor (e.g., touch sensitive sensor). In another implementation, the user interface **184K** is a sliding switch (e.g., sliding lever). In another implementation, the user interface **184K** is a rotatable dial. Advantageously, actuation of the user interface **184K** can alter the information

shown on the display screen **180K**, such as the form of a shipping label shown on an E-ink display screen **180K**. For example, actuation of the user interface **184K**, can switch the text associated with the sender and receiver, allowing the container system **100K** to be shipped back to the sender once the receiving party is done with it.

[0135] FIG. **19B** shows a block diagram of electronics **500** of the container system **100K**. The electronics **500** can include circuitry EM' (e.g., including one or more processors on a printed circuit board). The circuitry EM' communicate with one or more batteries PS', with the display screen **180K**, and with the user interface **184K**. Optionally, a memory module **185K** is in communication with the circuitry EM'. In one implementation, the memory module **185K** can optionally be disposed on the same printed circuit board as other components of the circuitry EM'. The circuitry EM' optionally controls the information displayed on the display screen **180K**. Information (e.g., sender address, recipient address, etc.) can be communicated to the circuitry EM' via an input module **186K**. The input module **186K** can receive such information wirelessly (e.g., via radiofrequency or RF communication, via infrared or IR communication, via WiFi 802.11, via BLUETOOTH®, etc.), such as using a wand (e.g., a radiofrequency or RF wand that is waved over the container system **100K**, such as over the display screen **180K**, where the wand is connected to a computer system where the shipping information is contained). Once received by the input module **186K**, the information (e.g., shipping information for a shipping label) to be displayed on the display screen **180K** can be electronically saved in the memory module **185K**. Advantageously, the one or more batteries PS' can power the electronics **500**, and therefore the display screen **180K** for a plurality of uses of the container **100K** (e.g., during shipping of the container system **100K** up to one-thousand times).

[0136] FIG. **20A** shows a block diagram of one method **700A** for shipping the container system **100K**. At step **710**, one or more containers, such as containers **150**, **150J** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, vaccines, medicine such as insulin, epinephrine, etc.) are placed in the container vessel **120K** of the container system **100K**, such as at a distribution facility for the containers **150**, **150J**. At step **720**, the lid L is closed over the container vessel **120K** once finished loading all containers **150**, **150J** into the container vessel **120K**. Optionally, the lid L is locked to the container vessel **120K** (e.g., via a magnetically actuated lock, including an electromagnet actuated when the lid is closed that can be turned off with a code, such as a digital code). At step **730**, information (e.g., shipping label information) is communicated to the container system **100K**. For example, as discussed above, a radiofrequency (RF) wand can be waved over the container system **100K** (e.g., over the lid L) to transfer the shipping information to the input module **186K** of the electronics **500** of the container system **100K**. At step **740**, the container system **100K** is shipped to the recipient (e.g., displayed on the shipping label **182K** on the display screen **180K**).

[0137] FIG. **20B** shows a block diagram of a method **700B** for returning the container **100K**. At step **750**, after receiving the container system **100K**, the lid L can be opened relative to the container vessel **120K**. Optionally, prior to opening the lid L, the lid L is unlocked relative to the container vessel **100K** (e.g., using a code, such as a digital code, provided to the recipient from the shipper). At step **760**, the one or more containers **150**, **150J** are removed from the container vessel **120K**. At step **770**, the lid L is closed over the container vessel **120K**. At step **780**, the user interface **184K** (e.g., button) is actuated to switch the information of the sender and recipient in the display screen **180** with each other, advantageously allowing the return of the container system **100K** to the original sender to be used again without having to reenter shipping information on the display screen **180K**. The display screen **180K** and label **182K** advantageously facilitate the shipping of the container system **100K** without having to print any separate labels for the container system **100K**. Further, the display screen **180K** and user interface **184K** advantageously facilitate return of the container system **100K** to the sender (e.g. without having to reenter shipping information, without having to print any labels), where the container system **100K** can be reused to ship containers **150**,

150J (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, vaccines, medicine such as insulin, epinephrine, etc.) again, such as to the same or a different recipient. The reuse of the container system **100K** for delivery of perishable material (e.g., medicine) advantageously reduces the cost of shipping by allowing the reuse of the container vessel **120K** (e.g., as compared to commonly used cardboard containers, which are disposed of after one use).

[0138] FIGS. **21A-21D** show different screens of a graphical user interface (GUI) used on a remote electronic device (e.g., mobile electronic device, such as a mobile phone, tablet computer). The GUI advantageously allows a user to interface with the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L** provide control settings (e.g., temperature presets for different medications in the containers **150**, **150J**), provide scheduling information (e.g., for the consumption of medication in the containers **150**, **150J**), provide alerts (e.g., battery life of the cooling system, temperature of the container(s) **150**, **150J**). The GUI can provide additional information not shown on the screens in FIGS. **21A-21D**. Via the GUI, a user can communicate with the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L** when they are ready to ingest the contents of the container **150**, **150J** and the system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L** can optionally heat one of the containers **150**, **150J** a predetermined temperature (e.g., body temperature, room temperature) and optionally alert the user when ready (via the GUI) to notify the user when the contents (e.g., medication) is ready for consumption. Optionally, where the container vessel **120**, **120E**, **120F**, **120G**, **120H**, **120I**, **120J**, **120K**, **120L** includes more than one container **150**, **150J**, the user can communicate via the GUI with the system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L** to prepare (e.g., heat) one of the containers (e.g., to body temperature) while the rest of the containers **150**, **150J** in the container vessel **100** remain in a cooled state. Optionally, once the container **150**, **150J** has been prepared (e.g., heated), in addition to notifying the user that the contents (e.g., medication) in the container **150**, **150J** is ready for consumption, it can also actuate the chamber **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126L** to move it to the extended position (e.g., via one of the linear actuation mechanisms disclosed herein) so when the user removes the lid from the container vessel **120**, **120E**, **120F**, **120G**, **120H**, **120I**, **120J**, **120K**, **120L** the user can readily identify which of the containers **150**, **150J** is the one that is ready for consumption (e.g., which one has been heated to room temperature or body temperature), while the rest of the chambers **126**, **126'**, **126E**, **126F1**, **126F2**, **126G1**, **126L** remain in the retracted position. [0139] FIGS. **22A-22B** show a container system **100L** (e.g., capsule container) that includes a cooling system **200L**. Some of the features of the container system **100L** and cooling system **200L** are similar to features of the container system **100-100K** and cooling system **200-200K** in FIGS. **1-19A**. Thus, reference numerals used to designate the various components of the container system **100L** and cooling system **200L** are identical to those used for identifying the corresponding components of the container system **100-100K** and cooling system **200-200K** in FIGS. **1-19A**, except that an "L" has been added to the numerical identifier. Therefore, the structure and description for the various features of the container system **100-100K** and cooling system **200-200K** and how it's operated and controlled in FIGS. **1-19A** are understood to apply to the corresponding features of the container system **100L** and cooling system **200L** in FIGS. **22A-22B**, except as described below.

[0140] The container system **100L** has a container vessel **120L** that is optionally cylindrical. The container vessel **120L** is optionally a cooler with active temperature control provided by the cooling system **200L** to cool the contents of the container vessel **120L** and/or maintain the contents of the vessel **120L** in a cooled or chilled state. Optionally, the vessel **120L** can hold therein one or more (e.g., a plurality of) separate containers **150** (e.g., medicine containers, such as injector pens, vials, cartridges (such as for injector pens), etc.). Optionally, the one or more (e.g., plurality of) separate containers **150** that can be inserted into the container vessel **120L** can contain a medication or medicine (e.g., epinephrine, insulin, vaccines, etc.).

[0141] The container vessel **120L** has an outer wall **121L** that extends between a proximal end **122L** that has an opening and a distal end **124L** having a base **125L**. The opening is selectively closed by a lid **L** removably attached to the proximal end **122L**. the vessel **120L** has an inner wall **126AL** and a base wall **126BL** that together define an open chamber **126L** that can receive and hold contents to be cooled therein (e.g., medicine containers, such as one or more vials, cartridges, injector pens, etc.). The vessel **120L** can optionally have an intermediate wall **126CL** spaced about the inner wall **126AL** and base wall **126BL**, such that the intermediate wall **126CL** is at least partially disposed between the outer wall **121L** and the inner wall **126AL**. The intermediate wall **126CL** is spaced apart from the inner wall **126AL** and base wall **126BL** so as to define a gap **128L** between the intermediate wall **126CL** and the inner wall **126AL** and base wall **126BL**. The gap **128L** can optionally be under vacuum so that the inner wall **126AL** and base wall **126BL** are vacuum insulated relative to the intermediate wall **126CL** and the outer wall **121L** of the vessel **120L**.

[0142] Optionally, one or more of the inner wall **126AL**, intermediate wall **126CL** and outer wall **121L** can be made of metal (e.g., stainless steel). In one implementation, the inner wall **126AL**, base wall **126BL** and intermediate wall **126CL** are made of metal (e.g., stainless steel). In another implementation, one or more portions (e.g., outer wall **121L**, intermediate wall **126CL** and/or inner wall **126AL**) of the vessel **120L** can be made of plastic.

[0143] The vessel **120L** has a cavity **127L** between the base wall **126BL** and the base **125L** of the vessel **120L**. The cavity **127L** can optionally house electronics, such as, for example, one or more batteries **277L** and one or more printed circuit boards (PCBA) with circuitry that controls the operation of the cooling system **200L**. In one implementation, the cavity **127L** can optionally house a power button or switch actuatable by a user through the bottom of the vessel **200L**. Optionally, at least a portion of the base **125L** (e.g. a cap of the base **125L**) is removable to access the electronics in the cavity **127L** (e.g., to replace the one or more batteries **277L**, perform maintenance on the electronics, such as the PCBA, etc.). The power button or switch is accessible by a user (e.g., can be pressed to turn on the cooling system **200L**, pressed to turn off the cooling system **200L**, pressed to pair the cooling system **200L** with a mobile electronic device, etc.). Optionally, the power switch can be located generally at the center of the base **125L**.

[0144] The cooling system **200L** is optionally at least partially housed in the vessel **120L**. In one implementation, the cooling system **200L** can include a first heat sink (cold side heat sink) **210L** in thermal communication with one or more thermoelectric elements (TECs) **220L**, such as Peltier element(s), and can be in thermal communication with the chamber **126L** of the vessel **120L** (e.g., via contact with the inner wall **126AL**, via conduction with air in the chamber **126L**, etc.). The first heat sink **210L** portion outside the vessel **120L** communicates with the first heat sink **210L** portion inside the vessel **120L** via a first heat sink **210L** portion (e.g., bridge portion) that interconnects the portions of the first heat sink **210L** outside and inside the vessel **120L**.

[0145] The one or more TECs **220L** are selectively operated (e.g., by the circuitry) to draw heat from the first heat sink (e.g., cold-side heat sink) **210L** and transfer it to the second heat sink (hot-side heat sink) **230L**. A fan **280L** is selectively operable to draw air into the vessel **120L** (e.g., into a channel **FP** of the vessel **120L**) to dissipate heat from the second heat sink **230L**, thereby allowing the TECs **220L** to draw further heat from the first heat sink **210L**, and thereby draw heat from the chamber **126L**. During operation of the fan **280L**, intake air flow F_i is drawn through one or more intake vents **203L** (having one or more openings) in the vessel **120L** and over the second heat sink **230L** (where the air flow removes heat from the second heat sink **230L**), after which the exhaust air flow F_o flows out of one or more exhaust vents **205L** (having one or more openings) in the vessel **120L**.

[0146] The chamber **126L** optionally receives and holds one or more (e.g., a plurality of) containers **150** (e.g., medicine containers, such as injector pens or cartridges for injector pens, vials, etc.). In one implementation, the first heat sink **210L** can be made of aluminum. However, the first heat sink **210L** can be made of other suitable materials (e.g., metals with high thermal conductivity).

[0147] The electronics (e.g., PCBA, batteries **277L**) can electrically communicate with the fan **280L** and TECs **220L**. Accordingly, power can be provided from the batteries **277L** to the TECs **220L** and/or fan **280L**, and the circuitry (e.g., in or on the PCBA) can control the operation of the TECs **220L** and/or fan **280L**.

[0148] The container **100L** can optionally have a visual display on the outer wall **121L** of the vessel **120L** (e.g., on the lid **L**). The visual display can optionally display one or more of the temperature in the chamber **126L**, the temperature of the first heat sink **210L**, the ambient temperature, a charge level or percentage for the one or more batteries **277L**, and amount of time left before recharging of the batteries **277L** is needed, etc. The visual display can optionally include a user interface (e.g., pressure sensitive buttons, capacitance touch buttons, etc.) to adjust (up or down) the temperature preset at which the cooling system **200L** is to cool the chamber **126L**. Accordingly, the operation of the container **100L** (e.g., of the cooling system **200L**) can be selected via the visual display and user interface on a surface of the container **100L**. Optionally, the visual display can include one or more hidden-til-lit LEDs. Optionally, the visual display can include an electrophoretic or electronic ink (e-ink) display. In one variation, the container **100L** can optionally include a hidden-til-lit LED that can selectively illuminate (e.g., to indicate one or more operating functions of the container **100L**, such as to indicate that the cooling system **200L** is in operation). The LED can optionally be a multi-color LED selectively operable to indicate one or more operating conditions of the container **100L** (e.g., green if normal operation, red if abnormal operation, such as low battery charge or inadequate cooling for sensed ambient temperature, etc.).

[0149] In operation, the cooling system **200L** can optionally be actuated by pressing a power button. Optionally, the cooling system **200L** can additionally (or alternatively) be actuated remotely (e.g., wirelessly) via a remote electronic device, such as a mobile phone, tablet computer, laptop computer, etc. that wirelessly communicates with the cooling system **200L** (e.g., with a receiver or transceiver of the circuitry). In still another implementation, the cooling system **200L** can automatically cool the chamber **126L** when the lid **L** is in a closed position on the vessel **120L**. The chamber **126L** can be cooled to a predetermined and/or a user selected temperature or temperature range, or automatically cooled to a temperature preset corresponding to the contents in the containers **150** (e.g., insulin, epinephrine, vaccines, etc.). The user selected temperature or temperature range can be selected via a user interface on the container **100L** and/or via the remote electronic device.

[0150] In one variation, the container system **100L** is powered using 12 VDC power (e.g., from one or more batteries **277L** or a power base on which the vessel **120L** is placed). In another variation, the container system **100L** is powered using 120 VAC or 240 VAC power, for example using a power base. The circuitry in the container **100L** can include a surge protector to inhibit damage to the electronics in the container **100L** from a power surge. The container system **100L** is advantageously easy to assemble and simpler to use. For example, inclusion of the cooling system **200** in the vessel **120L** makes it easier for users with limitations in hand articulation (e.g., users suffering from arthritis) to open the lid **L** (e.g., because it is lighter or weighs less) to remove the container(s) **150** (e.g., vaccines, insulin, medical containers) from the chamber **126L**. The lid **L** can optionally be insulated (e.g., be made of a hollow plastic body filled with foam insulation, such as light density Styrofoam).

[0151] While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. For example, though the features disclosed herein are described for medicine containers, the features are applicable to containers that are not medicine containers (e.g., portable coolers for food, chilled water cooler/bottle, etc.) and the invention is understood to extend to such other containers. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The

accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

[0152] Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

[0153] Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

[0154] Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

[0155] For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

[0156] Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

[0157] Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

[0158] Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

[0159] The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

Claims

1. A portable cooler container with active temperature control for one or more injector pens, comprising: a container vessel having an outer wall that extends between a proximal end that has an opening and a distal end having a base, the opening selectively closed by a lid, the container vessel having a chamber sized to receive one or more injector pens, a chamber wall being spaced inward of an inner wall to define a gap between the chamber wall and the inner wall, the gap filled with a phase change material; and a temperature control system comprising: one or more thermoelectric elements configured to heat or cool at least a portion of the chamber, a heat sink in thermal communication with one side of the one or more thermoelectric elements and in thermal communication with the phase change material, one or more power storage elements, and circuitry configured to control the one or more thermoelectric elements to heat or cool at least the portion of the chamber to a predetermined temperature or temperature range, wherein the one or more thermoelectric elements are operable to heat or cool the phase change material.
2. The container of claim 1, wherein a second heat sink is in thermal communication with an opposite side of the one or more thermoelectric elements.
3. The container of claim 1, further comprising a user interface configured to display information indicative of a temperature in the chamber, an ambient temperature, or a charge level of the one or more power storage elements.
4. The container of claim 1, wherein the chamber comprises two spaced apart chambers, each chamber configured to receive one injector pen.
5. The container of claim 2, wherein one or more fans are operable to draw air through one or more air intake vents, to flow said air over the second heat sink to dissipate heat from the second heat sink, and to then flow said air through one or more exhaust vents.
6. The container of claim 1, wherein the chamber wall is cylindrical.
7. The container of claim 1, further comprising a power base configured to receive the container vessel thereon and to provide power to the one or more thermoelectric elements or one or more power storage elements via inductive coupling.
8. A portable cooler container with active temperature control for one or more injector pens,

comprising: a container vessel having an outer wall that extends between a proximal end that has an opening and a distal end having a base, the opening selectively closed by a lid, the container vessel having a chamber sized to receive one or more injector pens, a chamber wall being spaced inward of an inner wall to define a gap between the chamber wall and the inner wall, the gap filled with a phase change material; and a temperature control system comprising: one or more thermoelectric elements configured to heat or cool at least a portion of the chamber, a heat sink in thermal communication with one side of the one or more thermoelectric elements and in thermal communication with the phase change material, one or more power storage elements, circuitry configured to control the one or more thermoelectric elements to heat or cool at least the portion of the chamber to a predetermined temperature or temperature range; and one or more sensors configured to sense one or more parameters of the chamber and to communicate the sensed parameters to the circuitry, wherein the one or more thermoelectric elements are operable to heat or cool the phase change material.

9. The container of claim 8, wherein a second heat sink is in thermal communication with an opposite side of the one or more thermoelectric elements.

10. The container of claim 8, further comprising a user interface configured to display information indicative of a temperature in the chamber, an ambient temperature, or a charge level of the one or more power storage elements.

11. The container of claim 8, wherein the chamber comprises two spaced apart chambers, each chamber configured to receive one injector pen.

12. The container of claim 8, wherein the chamber wall is cylindrical.

13. The container of claim 8, further comprising a power base configured to receive the container vessel thereon and to provide power to the one or more thermoelectric elements or one or more power storage elements via inductive coupling.

14. A portable cooler container with active temperature control for one or more injector pens, comprising: a container vessel having an outer wall that extends between a proximal end that has an opening and a distal end having a base, the opening selectively closed by a lid, the container vessel having a chamber sized to receive one or more injector pens, a chamber wall being spaced inward of an inner wall to define a gap between the chamber wall and the inner wall, the gap filled with a phase change material; and a temperature control system comprising: one or more thermoelectric elements, a heat sink in thermal communication with one side of the one or more thermoelectric elements and in thermal communication with the phase change material, one or more power storage elements, circuitry configured to control the one or more thermoelectric elements; and one or more sensors configured to sense one or more parameters of the chamber and to communicate the sensed parameters to the circuitry, wherein the one or more thermoelectric elements are operable to heat or cool the phase change material.

15. The container of claim 14, further comprising a user interface configured to display information indicative of a temperature in the chamber, an ambient temperature, or a charge level of the one or more power storage elements.

16. The container of claim 14, wherein the chamber comprises two spaced apart chambers, each chamber configured to receive one injector pen.

17. The container of claim 14, further comprising one or more light emitting diodes (LEDs) operable to selectively illuminate to indicate one or more operating functions of the container.

18. The container of claim 14, wherein the chamber wall is cylindrical.

19. The container of claim 14, further comprising a power base configured to receive the container vessel thereon and to provide power to the one or more thermoelectric elements or one or more power storage elements via inductive coupling.
